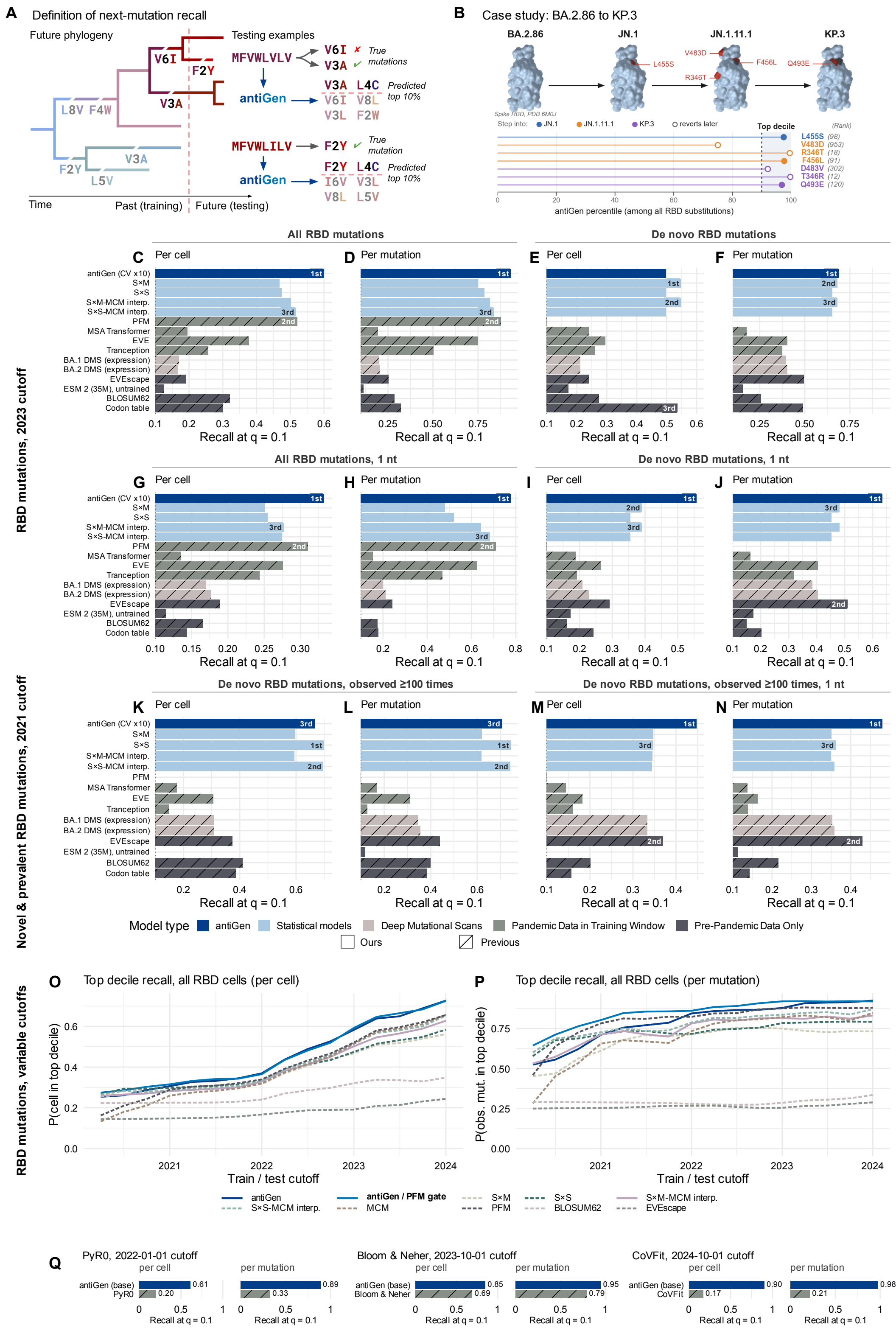

A
Definition of next-mutation recall
Future phylogeny
Testing examples
V6I
F2Y
V3A
L8V F4W
F2Y
V3A
L5V
MFVWLVLV
V6I ✗ True mutations
V3A ✓
antiGen
V3A L4C Predicted top 10%
V6I V8L
V3L F2W
MFVWLILV
F2Y ✓ True mutation
antiGen
F2Y L4C Predicted top 10%
I6V V3L
V8L L5V
Time
Past (training)
Future (testing)
B
Case study: BA.2.86 to KP.3
BA.2.86
JN.1
JN.1.11.1
KP.3
L455S
V483D
F456L
R346T
Q493E
Spike RBD, PDB 6M0J
Step into: JN.1 JN.1.11.1 KP.3 reverts later
Top decile
(Rank)
L455S (98)
V483D (953)
R346T (18)
F456L (91)
D483V (302)
T346R (12)
Q493E (120)
antiGen percentile (among all RBD substitutions)
All RBD mutations
De novo RBD mutations
C Per cell
D Per mutation
E Per cell
F Per mutation
antiGen (CV x10)
S×M
S×S
S×M-MCM interp.
S×S-MCM interp.
PFM
MSA Transformer
EVE
Tranception
BA.1 DMS (expression)
BA.2 DMS (expression)
EVEscape
ESM 2 (35M), untrained
BLOSUM62
Codon table
Recall at q = 0.1
All RBD mutations, 1 nt
De novo RBD mutations, 1 nt
G Per cell
H Per mutation
I Per cell
J Per mutation
RBD mutations, 2023 cutoff
De novo RBD mutations, observed ≥100 times
De novo RBD mutations, observed ≥100 times, 1 nt
K Per cell
L Per mutation
M Per cell
N Per mutation
Novel & prevalent RBD mutations, 2021 cutoff
Model type
antiGen
Statistical models
Deep Mutational Scans
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous
RBD mutations, variable cutoffs
O Top decile recall, all RBD cells (per cell)
P Top decile recall, all RBD cells (per mutation)
P(cell in top decile)
P(obs. mut. in top decile)
Train / test cutoff
antiGen
antiGen / PFM gate
S×M
S×S
S×M-MCM interp.
S×S-MCM interp.
MCM
PFM
BLOSUM62
EVEscape
Q
PyR0, 2022-01-01 cutoff
Bloom & Neher, 2023-10-01 cutoff
CoVFit, 2024-10-01 cutoff
per cell
per mutation
antiGen (base)
PyR0
Bloom & Neher
CoVFit
Recall at q = 0.1